

Lucas Starkey

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Education

Tennessee Technological University

Exp: Dec 2026

Major: Computer Science; Concentration: Data Science and Artificial Intelligence

GPA: Cookeville, TN

Technical Projects

Python, Graph Theory, Stochastic Modeling

Ticket to Ride Simulation & AI Framework

Architect & Solo Developer

- **Simulation Engine:** Engineered a high-fidelity game engine from scratch, implementing adjacency-list graph construction and CSV-driven data ingestion to enable modular, scalable map loading.
- **Graph Algorithms:** Developed a custom path-evaluation system using recursive depth-first search (DFS) to calculate connected components and recompute longest path metrics in $O(V+E)$ time during real-time state changes.
- **Security & Architecture:** Built a secure agent interface using defensive deepcopying and weak references to enforce hidden-information constraints, preventing memory leaks and state injection during competitive play.
- **AI & Decision Theory:** Built a probabilistic decision-making agent that maximizes expected utility using Bayesian inference to estimate opponent hidden states from observed actions, enabling predictive modeling, uncertainty quantification, and data-driven strategic optimization in a partially observable environment.
- **Data Interpretability:** Analyzed tournament execution logs to extract winning heuristics, translating complex agent behavior into interpretable strategies and actionable insights for human competitive play.

Work Experience

Feb 2025 - Current

Averitt Express

Tech Support Specialist

- Orchestrate weekly maintenance for hybrid IBM AS/400 and Windows production environments, utilizing Bash and CLI to ensure 99.9% system uptime.
- **Cross-Functional Data Integration:** Collaborated with three internal departments to audit monitoring requirements; engineered custom integrations to fetch and centralize siloed metrics into Grafana for real-time enterprise observability.
- Execute system IPL processes and manage Hardware Management Console (HMC) operations to minimize downtime during critical maintenance windows.
- Proactively monitor infrastructure health using VMware, Adeptia, and WhatsUp Gold to identify and resolve system bottlenecks before they impact operations.

Clubs and Organization

Jan 2024 - Current

Baptist Collegiate Ministry

Technical Leadership & Data Systems

- Engineered an end-to-end attendance and engagement tracking system to resolve critical visibility gaps in organizational growth metrics.
- Developed and deployed automated leader assessments and surveys; analyzed results to bridge the gap between "idealized" goals and "on-the-ground" reality.
- Leveraged survey data to overhaul the leadership interviewing process and design targeted training curricula, ensuring clear, data-backed expectations for over 45 small group leaders.
- Developed custom integration using Canva's REST API to automate slide aggregation, reducing weekly event setup time by 50%.

Skills

Languages & Frameworks: Python (NumPy, Pandas), R, C++, JavaScript, Bash, SQL, YAML, HTML/CSS.

Data & AI: Bayesian Inference, Expected Utility, Stochastic Modeling, Graph Theory, Grafana, RESTful APIs, JSON.

DevOps & Infrastructure: Docker, Git/GitHub, Linux Admin, VMware, IBM iSeries (AS/400).

Developer Tools: VS Code, PyCharm, Knox Manage, Adeptia, RDP.